

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
## $n.burnin
## [1] 1000
##
## $n.draw
## [1] 1200
##
## $thin
## [1] 1
##
## $pp_cutoff
## [1] 0.5
##
## $prior_gamma
## [1] 0.1 0.1 0.1 0.1 0.1
##
## $K_values
## [1] "1 / Examined number of clusters: 2" "2 / Examined number of clusters: 3"
## [3] "3 / Examined number of clusters: 4" "4 / Examined number of clusters: 5"
## [5] "5 / Examined number of clusters: 6" "6 / Examined number of clusters: 7"
## [7] "7 / Examined number of clusters: 8" "8 / Examined number of clusters: 9"
## [9] "9 / Examined number of clusters: 10"
##
## $sdev
## [1] 0.05
##
## $beta.var.scale
## [1] 0.8
```

Summary Statistics

```
--- Summary for K = 1 ---
Z.ar (Location: fit[[1]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2117  0.2433  0.2525  0.2521  0.2608  0.2917

beta.ar (Location: fit[[1]]$beta.ar):
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.09833 0.14500 0.14120 0.17667 0.86167
```

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Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[1]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1017 0.1483 0.1447 0.1800 0.8142
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.102302 0.031667 1.000000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.009167 0.077821 0.024167 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005833 0.015000 0.162805 0.119167 1.000000

--- Summary for K = 2 ---
Z.ar (Location: fit[[2]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.06833 0.09167 0.09833 0.09872 0.10583 0.13000

beta.ar (Location: fit[[2]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.05167 0.10083 0.10753 0.14417 0.91000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[2]]$gamma.ar):
  Modality 1 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0550	0.1050	0.1103	0.1483	0.6775

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0008333	0.0358333	0.4032150	1.0000000	1.0000000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.003333	0.239622	0.274375	1.000000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.08083	1.00000	0.68568	1.00000	1.00000

--- Summary for K = 3 ---

Z.ar (Location: fit[[3]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.005833	0.017500	0.020833	0.021361	0.025000	0.041667

beta.ar (Location: fit[[3]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03583	0.07750	0.08754	0.12083	0.87500

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[3]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.03833	0.08167	0.09113	0.12667	0.67167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9542	1.0000	0.7950	1.0000	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2250	0.1708	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0550	1.0000	0.6758	1.0000	1.0000

```

--- Summary for K = 4 ---
Z.ar (Location: fit[[4]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.003333 0.016667 0.022500 0.023108 0.028333 0.085000

```

```

beta.ar (Location: fit[[4]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.016667 0.05167 0.06590 0.09500 0.81250
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

gamma.ar (Location: fit[[4]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.01833 0.05500 0.06909 0.10000 0.59750
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.3783  1.0000  0.7168  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2499  0.5104  1.0000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.0000  0.7108  1.0000  0.7748  1.0000  1.0000

```

```

--- Summary for K = 5 ---
Z.ar (Location: fit[[5]]$Z.ar):
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.001667 0.008333 0.012500 0.012952 0.015833 0.069167

```

```

beta.ar (Location: fit[[5]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.016667 0.05000 0.06351 0.09083 0.81333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

```

```

Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[5]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.05500 0.06764 0.09833 0.60500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.4692  1.0000  0.7306  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.0000  0.2521  0.5354  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.7942  1.0000  0.7952  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar (Location: fit[[6]]$Z.ar):
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.003333 0.005000 0.004964 0.005833 0.034167

beta.ar (Location: fit[[6]]$beta.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01667 0.04833 0.06395 0.09083 0.80250
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar (Location: fit[[6]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01833 0.05333 0.06880 0.10000 0.54000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.0000  0.5383  0.5051  1.0000  1.0000
  Modality 3 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9875	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2762	0.7819	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	1.00	1.00	0.85	1.00	1.00

--- Summary for K = 7 ---

Z.ar (Location: fit[[7]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.001667	0.006667	0.010000	0.010491	0.013333	0.031667

beta.ar (Location: fit[[7]]\$beta.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0100	0.0375	0.0537	0.0775	0.7167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar (Location: fit[[7]]\$gamma.ar):

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01167	0.04167	0.05863	0.08667	0.51167

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.3814	0.9625	1.0000

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.7558	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.1944	0.0000	1.0000

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5183	1.0000	0.7301	1.0000	1.0000

--- Summary for K = 8 ---

Z.ar (Location: fit[[8]]\$Z.ar):

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00500	0.01333	0.01667	0.01784	0.02167	0.04583

beta.ar (Location: fit[[8]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.006667 0.030000 0.047719 0.068333 0.706667
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

gamma.ar (Location: fit[[8]]\$gamma.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.008333 0.035000 0.052684 0.078333 0.548333
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2305 0.4675 1.0000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.1358 1.0000 1.0000 0.9986 1.0000 1.0000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.00000 0.09793 0.00000 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.4983 0.4846 1.0000 1.0000

```

--- Summary for K = 9 ---

Z.ar (Location: fit[[9]]\$Z.ar):

```

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.003333 0.012500 0.017500 0.017908 0.021667 0.055000

```

beta.ar (Location: fit[[9]]\$beta.ar):

```

Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00500 0.02458 0.04071 0.05917 0.68167
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1      1      1      1      1      1

```

```

gamma.ar (Location: fit[[9]]$gamma.ar):
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00500 0.02833 0.04618 0.06833 0.51000
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2589  0.5350  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  1.0000  1.0000  0.9895  1.0000  1.0000
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.0441  0.0000  1.0000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.2887  0.5967  1.0000

```